



MCS3301 – Control Engineering I

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Phase Lead and Lag Compensation

- A closed loop system response may often be approximated as a second order transfer function with damping ratio

$$\zeta = \frac{\phi_m}{100}$$

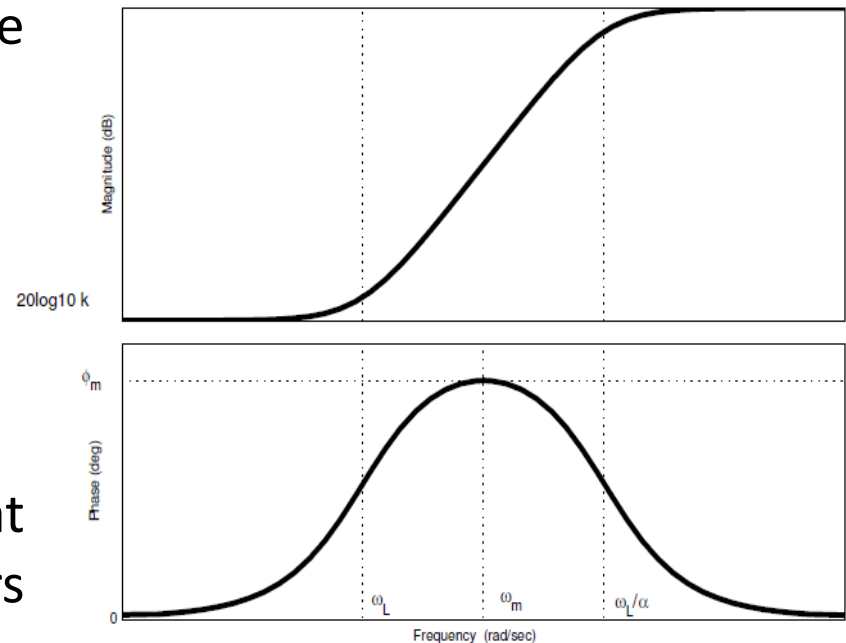
- Thus for a damping ratio of 0.5 or above, a phase margin of at least 50° is required.
- A **phase lead** compensator increases the phase margin but at the expense of increase gain.
- A phase lead compensator can be expressed as

$$C(s) = k \frac{1 + T_L s}{1 + \alpha T_L s} \quad \text{with } \{\alpha < 1\}$$

- Taking $\omega_L = 1/T_L$ then

$$C(s) = k \frac{s + \omega_L}{\alpha s + \omega_L}$$

- From the Bode plot it is observed that the maximum phase advance occurs when $\omega = \omega_m$.



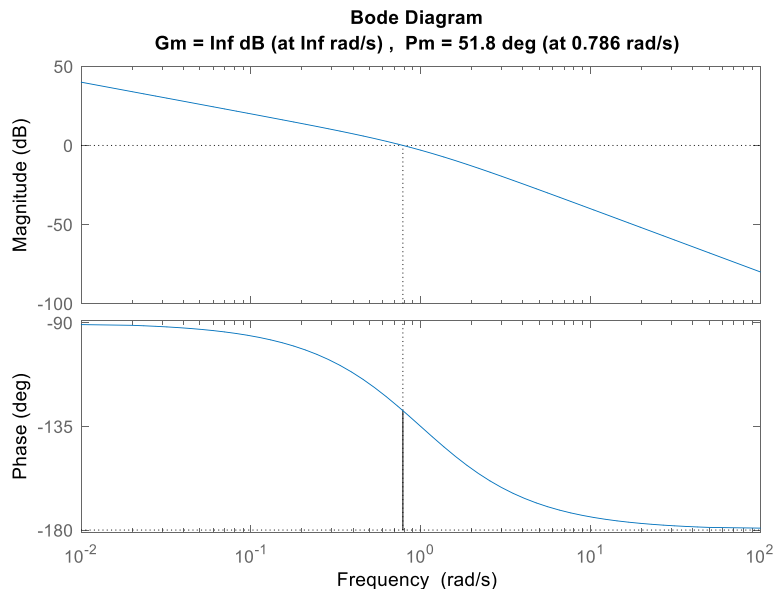
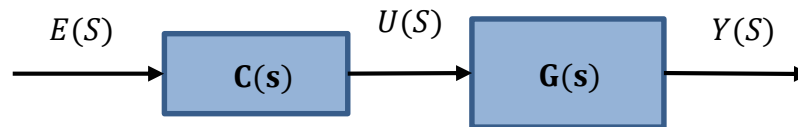


Phase Lead Compensation

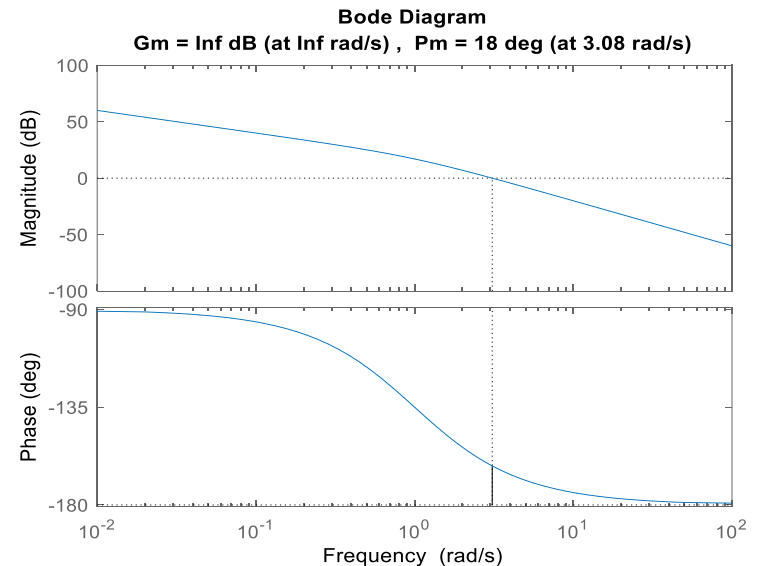
- The desired phase advance ϕ_m of a lead compensator can be specified as

$$\alpha = k \frac{1 - \sin \phi_m}{1 + \sin \phi_m}$$

- Consider the system below with $G(s) = \frac{1}{s(s+1)}$



$$C(s) = 1, GM = \infty \text{ and PM} = 51.8^\circ$$



$$C(s) = 10, GM = \infty \text{ and PM} = 18^\circ$$



Phase Lead Compensation

- Suppose we have a design specification that the cross-over frequency should be 3 rad/s but the phase margin should be at least 50°.
- A phase lead compensator with a maximum of 32° phase advance at 3 rad/s is thus needed.
- Let

$$\alpha = \frac{1 - \sin 32}{1 + \sin 32} \approx 0.31$$

$$\omega_l = \omega_m \sqrt{\alpha} = 1.6703$$

- Thus

$$C(s) = k \frac{\omega_l + s}{\omega_l + \alpha s} \approx k \frac{1.67 + s}{1.67 + 0.31s}$$

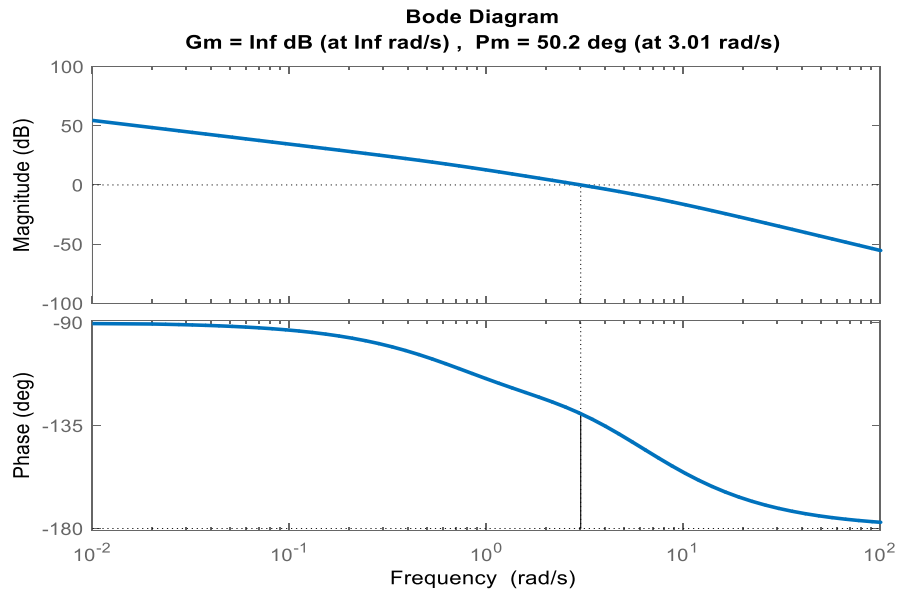
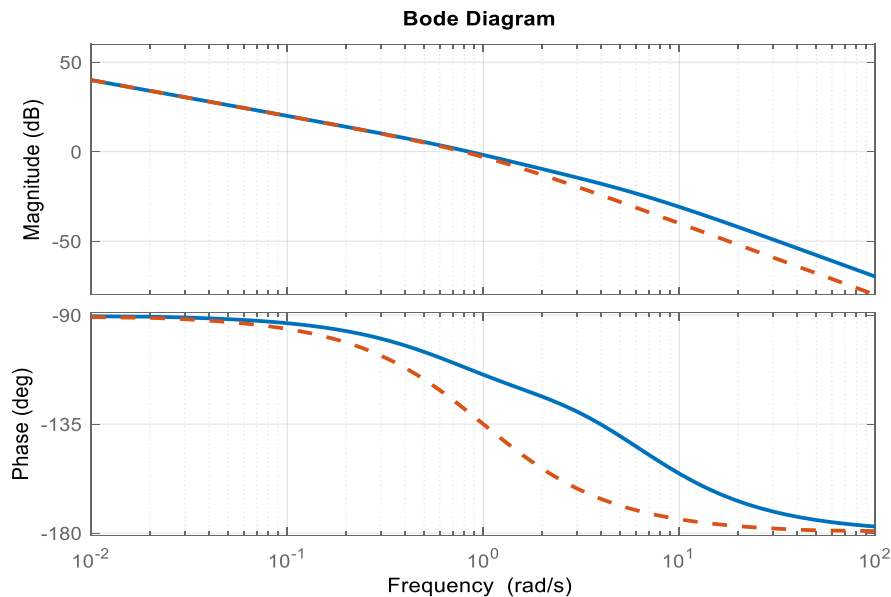
- The forward transfer function becomes

$$C(s)G(s) = k \frac{s + 1.67}{0.1s^3 + 1.98s^2 + 1.67s}$$



Phase Lead and Lag Compensation

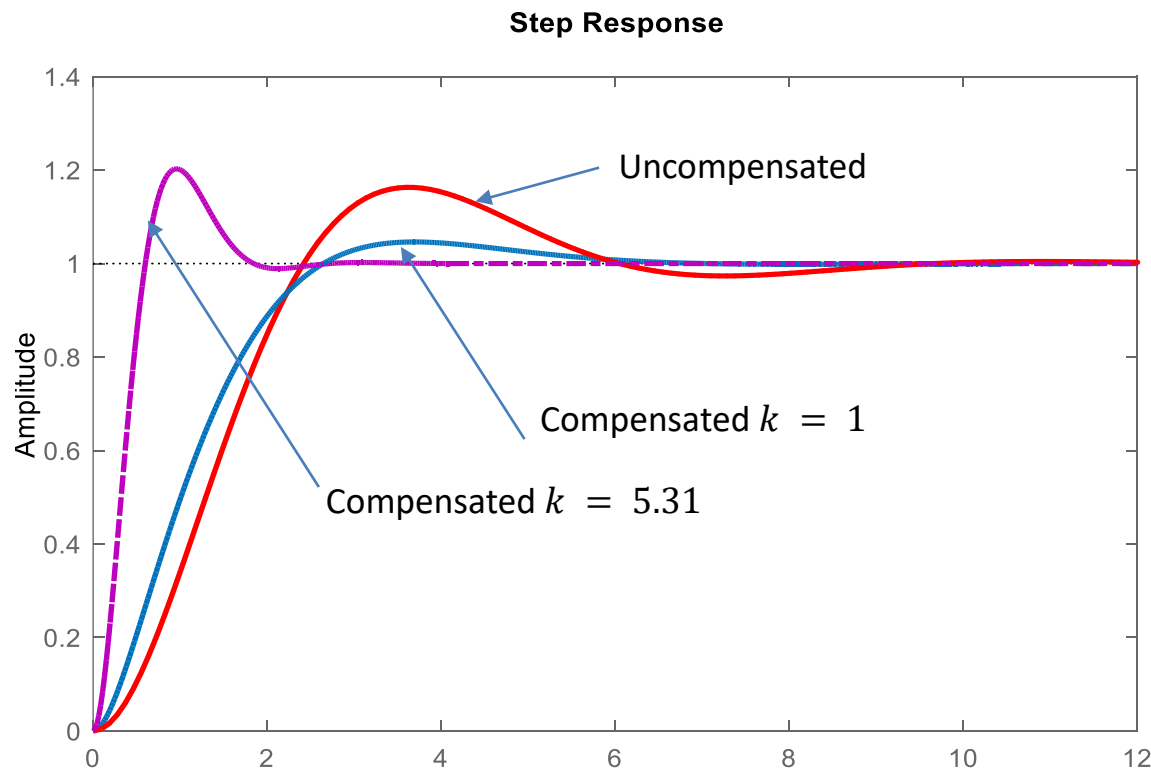
- The figure below shows the Bode plot of the uncompensated system $G(s)$ (dashed line) and the compensated (solid line) system with $k = 1$.
- To attain the cross over frequency of 3 rad/sec, k needs to be increased to 5.31.
- This can be deduced from the compensated system GC with $k = 1$ where the gain is -14.5 dB at $\omega = 3$.





Phase Lead Compensation: Time Domain

- The step responses of the uncompensated and lead compensated closed loop systems are shown below.
- Thus **phase lead** compensation can **increase damping** of a system (when used correctly).
- It improves the rise time and settling time.
- It does not affect the steady state error.





Phase Lag Compensation

- A phase lag compensator has the same structure with a phase lead compensator.

$$C(s) = k \frac{1 + T_L(s)}{1 + \alpha T_L(s)}$$

- However, in this case $\alpha > 0$.
- Its effect is opposite to that of a lead compensator
- That is, it gives extra gain at low frequencies at the expense of increased phase.
- Its role is thus similar to a Proportional Integral (PI) controller.
- If $\omega_L = 1/T_L$ then

$$C(s) = k \frac{s + \omega_L}{\alpha s + \omega_L}$$

- ω_L should be chosen well below the cross-over frequency so that the phase margin is not affected.



PID Control

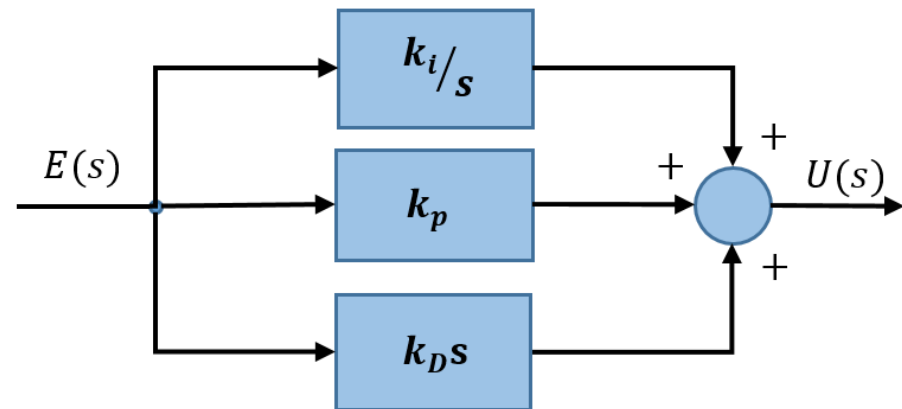
- PID stands for **Proportional-Integral-Derivative** Control.
- The output PID controller is defined by the equation

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau + k_d \frac{de}{dt}$$

- The Laplace transform of the above equations yields the transfer function

$$C(s) = \frac{U(s)}{E(s)} = k_p + \frac{k_i}{s} + k_d s$$

- Thus the output of the PID controller is a **linear** combination of the proportional, integral and derivative terms of the error.
- Owing to its relative simplicity, the PID controller is by far the most widely used feedback controller in the industry.





PID Control

- Proportional action affects the **current state** of the controller and there is always **steady state error** associated with proportional control.
- Integral action represents the **average of past errors** and helps to eliminate steady state error but can also lead to increased oscillatory behaviour.
- Derivative action can be interpreted as a **prediction of future errors** and helps to decrease oscillation but has the drawback of noise amplification.
- In the time domain, PID controllers are designed to have a fast **rise time**, fast **settling time**, minimum **overshoot** and minimum **Integral Error**.
- In the frequency domain, the PI controller is a low pass filter, the PD controller is a high pass filter.
- Thus the PID controller may be viewed as a **band-pass** filter.
- To achieve the desired results, the **PID gains** must be well tuned.



PID Control: Example

Example 1: Design a PI controller for the system $G(s)$ so that the closed loop system has damping ratio of 0.5 and natural frequency of 4 rad/sec. Find also the steady state gain of the system.

$$G(s) = \frac{2}{5s + 10}$$